

Field Maintenance Summary —

FMS-2025-005

Document ID: FMS-2025-005
Reporting Period: 2025-12-01 to 2025-12-31
Site / Region: NW-PowerPool-WIND-003 (primary); secondary references WIND-024
Field Supervisor: Aiko Tanaka-Pereira, Field Supervisor, NW-PowerPool-WIND-003
Distribution: Regional Maintenance Manager (Karen Hsu); Lead Field Supervisor (D. Mendoza-Ruiz); Operations Planning; Engineering Liaison; VP Operations (M. Ostrowski); Safety Officer

Section 1 — Reporting Period Summary

I am writing this report on the last working day of the year because I want it filed before the holiday standdown. The headline event for December at WIND-003 is the TRA-10561 transformer outage on 12/10. The headline is that it is the same transformer, in the same vendor's hands, doing the same emergency repair, seven months after the first one.

WO statistics, December 2025, NW-PowerPool-WIND-003:

Category	Planned	Completed	Deferred / Bumped	Emergency
Preventive WOs	11	7	4	n/a
Corrective WOs	8	5	3	n/a
Emergency WOs	0	n/a	n/a	2
Inspections	6	6	0	n/a

Section 2 — Significant Events

12/2 — WIN-10011 efficiency decline outage, NW-PowerPool-WIND-024 (cross-site reference).

Diego's site. Following on from the 11/23 pre-cursor alert in FMS-2025-004. Duration 111 hours. Lost generation 2,706.6 MWh. Sixth Apex deployment to WIND-024 in 12 months. I am noting it here for continuity even though the primary write-up will be OIR-2025-009.

12/10, 06:42 PT — TRA-10561 overheating outage, NW-PowerPool-WIND-003.

This is the event that prompted me to write the next paragraph carefully.

TRA-10561 is the Siemens transformer at the WIND-003 substation. It had a vibration-excess emergency outage on 2025-05-26 — 158 hours, 2,615.1 MWh lost, Apex on site with Siemens Gamesa OEM subcontractors, \$446,000 invoiced. That event is documented in OIR-2025-007 and the repair-vs-replace decision was deferred in RAM-2025-001.

On 2025-12-10, at 06:42 PT, the same transformer tripped on bushing overheating. Cooling fan circulation flow loss combined with elevated bushing temperature triggered the protection. Duration 215 hours. Lost generation 3,558.6 MWh. Will be written up as OIR-2025-004 per Engineering's numbering of the year-end events.

Apex was dispatched. The dispatch was the same crew lead, Brad Holstein. The dispatch repaired the same component family (bushings and cooling auxiliary). The invoice has not closed but the field estimate is \$670,000–\$750,000.

The same vendor is doing the same emergency repair on the same transformer. I want that sentence in the record because that is what is happening on the ground. The May repair did not address the underlying condition; it addressed the immediate failure mode. RAM-2025-001 acknowledged the repair-vs-replace question and deferred it. RAM-2025-004 has now been opened to escalate the same question for the second time on the same unit. I recommend immediate engineering review — and I mean the kind that produces a written disposition, not another deferral.

Section 3 — Vendor Activity

- **Apex:** Two dispatches in December at WIND-003 (TRA-10561 on 12/10, and a follow-up site visit 12/18 to verify cooling auxiliary). Combined estimated \$812,000 (not all invoices closed).
- **Apex (cross-site, WIND-024):** WIN-10011 emergency response 12/2. Estimated \$640,000.
- **Siemens Energy OEM warranty service:** Reviewed TRA-10561 on 12/15. Their finding was that the bushing replacement done in May "did not include the recommended desiccant cartridge replacement specified in OEM Manual Rev. 3.1, section 8.4." I am not certain that is the whole story — our copy of OEM Manual Rev. 3.1 section 8.4 references an ambient temperature de-rating practice that conflicts with NW-PowerPool's operating profile, and the field crew in May acted on what we believed was a reasonable substitute. There is a documentation conflict here that Engineering needs to resolve.
- **CPS:** Seven scheduled WOs, plus a precautionary visit to WIN-10085 on 12/8 (cooling-loop verification, no findings). Quiet month.

Section 4 — Open Issues / Backlog

- **TRA-10561** — second emergency in seven months. RAM-2025-004 escalation filed. Engineering review requested. The cost of two Apex emergencies on this unit (\$446K + estimated \$670K = \$1.12M) is approaching the cost of the full transformer replacement quote referenced in OER-2025-002 (\$2.4M).
 - **Siemens OEM Manual Rev. 3.1 vs. NW-PowerPool ambient practice** — documentation conflict, needs Engineering disposition. This is a recurring theme across our OEM base.
 - **WIN-10085** cooling-loop refurbishment is holding. Knock on something wooden.
 - **CSM-7 cadence dispute** — still unresolved. The TRA-10561 December event is going to trigger a procedural review whether anyone wants one or not.
 - **MOC-2025-022** — nine months.
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Section 5 — Resource & Safety Notes

Crew of 7 at WIND-003 (smaller site than WIND-024). One CPS technician on rotation. No FMP-2 exceedances in December but the holiday schedule is tight — I have one technician on call for 12/24–12/26 and one for 12/31–1/2.

Weather: cold, two ice days (12/12, 12/13). Restricted climbing on both. The TRA-10561 response happened in below-freezing conditions, which the crew handled within procedure.

No injuries. One near-miss reported on the TRA-10561 response (technician slipped on transformer cooling-fan platform; caught the rail). Logged.

Section 6 — Forward Look

January 2026 starts with two open questions on the table:

1. Will the TRA-10561 RAM-2025-004 escalation actually produce a determination, or will it be deferred a third time? Field judgment: if we patch and return to service again, we will see this transformer fail a third time before April.
2. Will the FY2026 capital plan include the WIND-024 aging-asset replacement line item that Margaret has been signaling? Diego and I have both written the same thing in different ways this fall: the current vendor-emergency-repair pattern is not a maintenance program.

I will be on rotation at WIND-024 in January and Diego will be back at WIND-003. The CPS surge extension request through Q1 is still unanswered.

Closing this report and the year. Wishing the field crew a quiet two weeks.

Submitted by:

Aiko Tanaka-Pereira
Field Supervisor, NW-PowerPool-WIND-003
Cascadia Renewable Operations LLC

References:

- OIR-2025-007 — TRA-10561 vibration excess outage, 2025-05-26 (Period A)
- OIR-2025-004 — TRA-10561 overheating outage, 2025-12-10 (Period B) — to be written
- RAM-2025-001 — TRA-10561 repair-vs-replace decision, May 2025 (deferred)
- RAM-2025-004 — TRA-10561 repair-vs-replace re-escalation, December 2025
- OER-2025-002 — Year-End FY2025 Western Region Risk Assessment
- FMS-2025-004 — November 2025, Diego's report on the emergency-repair cycle
- MOC-2025-022 — OEM Threshold Override (pending)